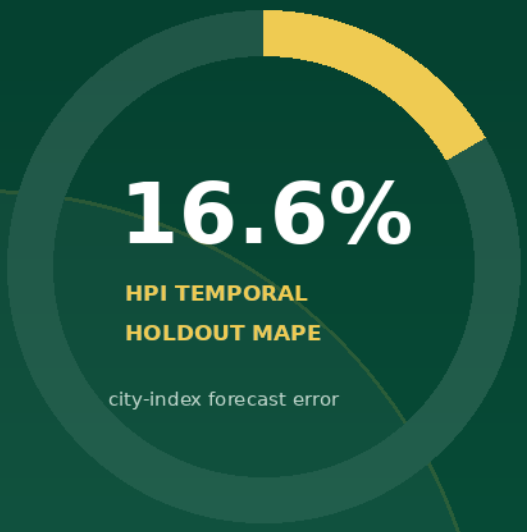
REAL ESTATE INVESTMENT ADVISOR

INDIA HOUSING MARKET / INVESTMENT ANALYTICS

Executive Report

Market structure, price signals,
forecast evaluation and decision readiness



250,000
LISTINGS

42
CITIES

28
HPI CITIES

20 Q
FORECAST HORIZON



PERFORMANCE & MARKET ANALYTICS

250,000 housing listings | NHB RESIDEX city index | Five-year scenario

Prepared 30 October 2026

Exploratory analysis only. Not financial advice or a property valuation.

Report at a glance

REPORT SCOPE	DETAIL
Dataset	India housing prices listing dataset (project attribution)
Listing coverage	250,000 cleaned records · 20 states · 42 cities
Market index	NHB RESIDEX HPI · 28 cities · 51 quarters (Jun 2013-Dec 2025)
Forecast horizon	20 quarters (five years), through Dec 2030
Evidence basis	Descriptive listing analysis, proxy-label diagnostics and city-level HPI forecast evaluation.

EXECUTIVE READOUT The project is a useful exploratory analytics prototype: it describes the supplied listing mix and market-price patterns, and evaluates a five-year city HPI scenario. The investment label is assumption-driven and the classifier ROC AUC is 0.501. Do not use the current model to rank properties or make investment decisions.

Prepared from project data, current analysis outputs and the supplied chart set. Listing prices are asking prices; NHB HPI is a separate city-level index.

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1. Executive summary

The project combines 250,000 Indian housing listings with NHB RESIDEX city assessment-price indices. It supports descriptive listing comparisons, a transparent age-based screening proxy and a five-year city-index scenario. The evidence is useful for analysis and product prototyping, but it does not establish realized investment returns or property-specific resale accuracy.

250,000 **CLEAN LISTINGS** Records retained in listing dataset	**20** **STATES REPRESENTED** Geographic coverage in listing data	**42** **CITIES REPRESENTED** City coverage in listing data
₹253.9 L **MEDIAN ASKING PRICE** Median listed property price	**₹9,245** **MEDIAN PRICE / SQ FT** Median listed price intensity	**2,747 sq ft** **MEDIAN PROPERTY AREA** Median listed property size

Table 1. Headline model and market indicators

INDICATOR	REPORTED VALUE	EXECUTIVE INTERPRETATION
NHB HPI rolling-origin CV MAPE	15.1% · 12 origins / 336 forecasts	Used to select the Theta method.
NHB HPI temporal holdout MAPE	16.6% · 8 origins / 224 forecasts	Later-origin city-index evaluation; not property resale error.
Investment classifier ROC AUC	0.501	Near-random ranking against a constructed proxy label.
Investment classifier F1	0.293	Weak predictive performance; not suitable for property ranking.
Proxy price regression R^2	≈ 1.000	Formula-generated target; near-perfect fit is not resale validation.

What matters for decisions

- The supplied listing charts show broad descriptive patterns, but several category distributions are strikingly similar; validate source generation, aggregation and group definitions before drawing market conclusions.
- The classifier target is defined from assumed appreciation less an age adjustment. The perfect negative age/proxy correlation is therefore built into the formula.
- The selected Theta model improves holdout MAE by about 28.7% versus a last-value baseline, but the HPI result is a city-index forecast with material forecast error.
- Asking prices are not completed transactions. The listing dataset has no verified purchase, resale, rental or operating-cost outcomes.

RECOMMENDATION Preserve this as exploratory decision support. Prioritize transaction-linked resale outcomes, locality-level enrichment and time/geography-based validation before presenting property-level investment recommendations.

2. Dataset footprint and listing composition

The supplied listing dataset offers broad geographic segmentation. State counts describe the composition of this file, not housing-market share or transaction activity. The paired median-asking-price view compares the same selected states.

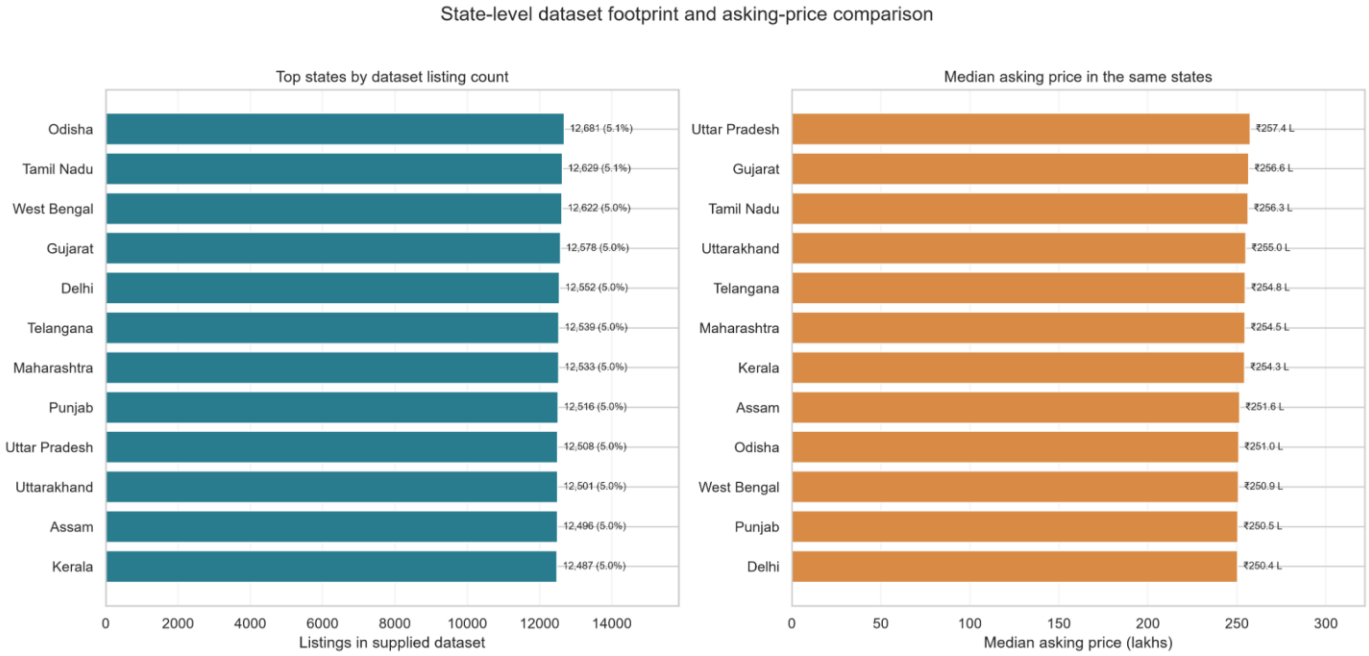


Figure 1. State-level dataset footprint and median asking-price comparison. Listing count is dataset coverage, not market share.

Listing profile

Availability, ownership and furnishing distributions provide context for the records represented in the file. They should not be interpreted as demand, vacancy, absorption or prevalence in the underlying housing market.

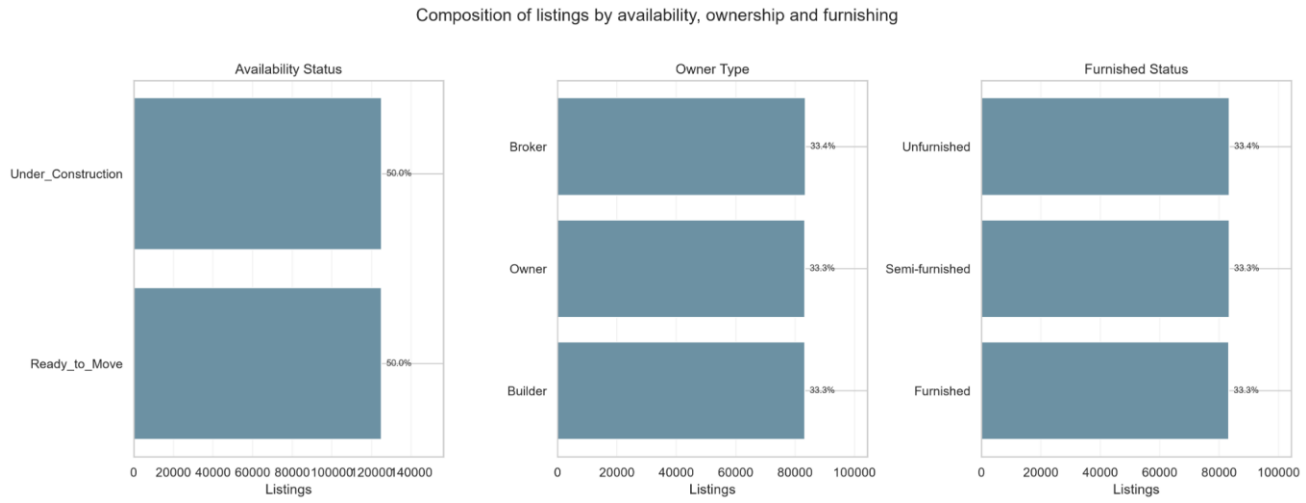


Figure 2. Listing composition across availability status, owner type and furnishing status.

READING THE CHART The supplied visual shows an even split between the two availability categories displayed and roughly one-third shares across the displayed owner and furnishing categories. Confirm full category coverage and counts from the exported summary tables before quoting exact proportions.

3. City-level market positioning

The city positioning view compares median asking price per square foot with median asking price. Bubble size represents the number of listing records; vertical and horizontal reference lines show the overall medians.



Figure 3. City positioning: median asking price per square foot versus median asking price. Bubble size indicates listing count.

Executive interpretation

- Cities in the upper-right quadrant are above the overall medians on both displayed asking-price measures; this is a descriptive shortlist for deeper investigation, not an investment ranking.
- The chart does not adjust for property type, BHK, area, age, locality or listing mix. These factors may affect city comparisons.
- Bubble size is record count in the supplied dataset, not transaction liquidity or market depth.

DECISION USE Use this chart to identify cities for a second-stage, like-for-like comparison. Validate minimum sample sizes and stratify by property type, BHK and area before making city recommendations.

4. City asking-price comparisons

The supplied top-20 city chart presents median listing prices in a relatively narrow range, approximately ₹248-260 lakh. The closeness of the displayed medians makes it especially important to validate the source distribution and city aggregation before describing meaningful price leaders.

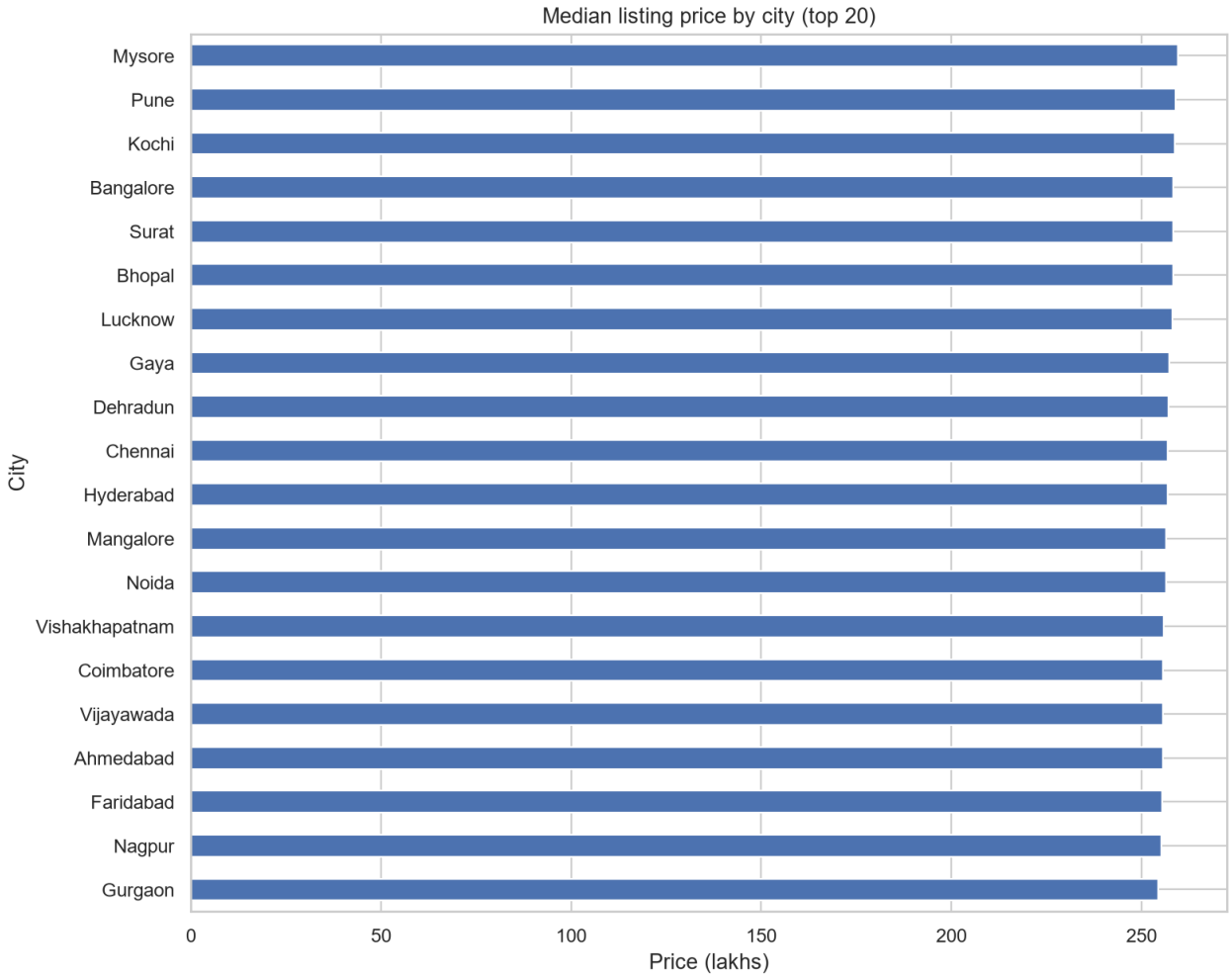


Figure 4. Median listing price by city (top 20) in the supplied chart output. Values are asking-price medians, not transaction benchmarks.

How to use this comparison

- Treat small differences between city medians as descriptive until confidence intervals and group sizes are reviewed.
- Compare like-for-like listings by property type, BHK and area to reduce mix effects.
- Do not infer affordability, appreciation or investment attractiveness from asking-price medians alone; income, rent, financing, transaction prices and holding costs are not present in this view.

DATA-QUALITY CHECK The tightly clustered values may reflect the supplied dataset structure. Recheck city-level aggregation, units, sample sizes and raw distributions before using this chart to rank markets.

5. Property attributes and price dispersion

The property-type chart compares asking-price distributions for villas, independent houses and apartments, with the display limited to the 1st-99th percentiles. The observed distributions are visually similar across the three groups.



Figure 5. Asking-price spread by property type; display limited to the 1st-99th percentiles. Sample counts are shown in category labels.

The age and furnishing summaries also show nearly identical medians and interquartile ranges in the supplied visualization. This does not prove those attributes are irrelevant; it signals that category definitions, group sizes, market mix and possible data-generation effects should be examined.

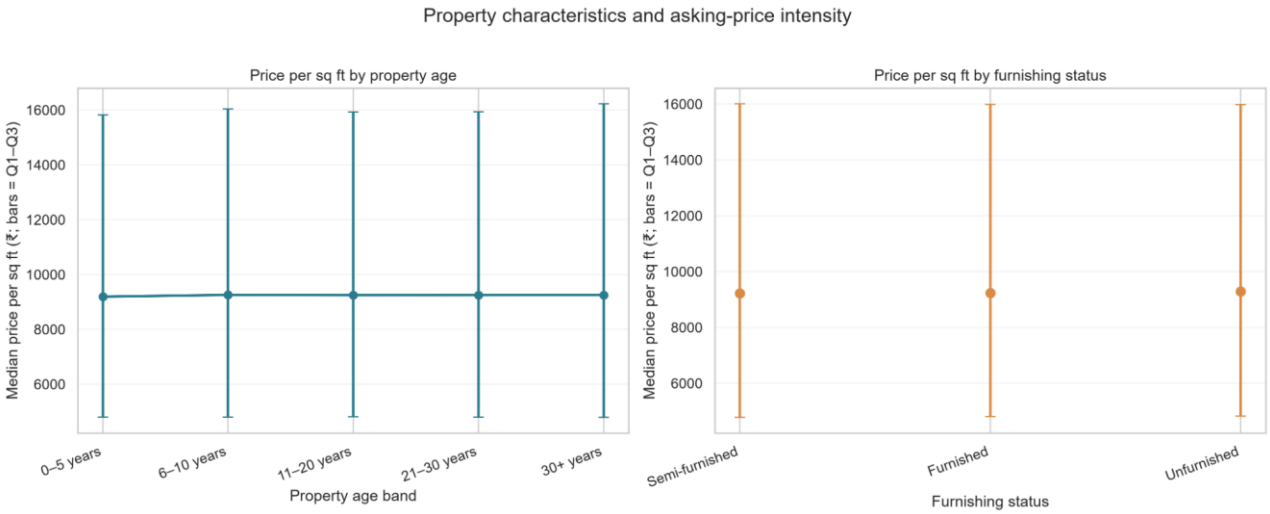


Figure 6. Median price per square foot and interquartile ranges by property-age band and furnishing status.

ANALYST INTERPRETATION Before drawing attribute-level conclusions, segment by city, BHK, size and property type, and confirm that the source contains sufficient independent variation across each attribute.

6. Proxy target and relationship diagnostics

The project's "Good Investment" target is a rule-generated label. It assumes 6% annual appreciation, subtracts 0.2 percentage points per property year, and marks values above 4% as positive. It is a transparent screening demonstration, not a measured investment outcome.

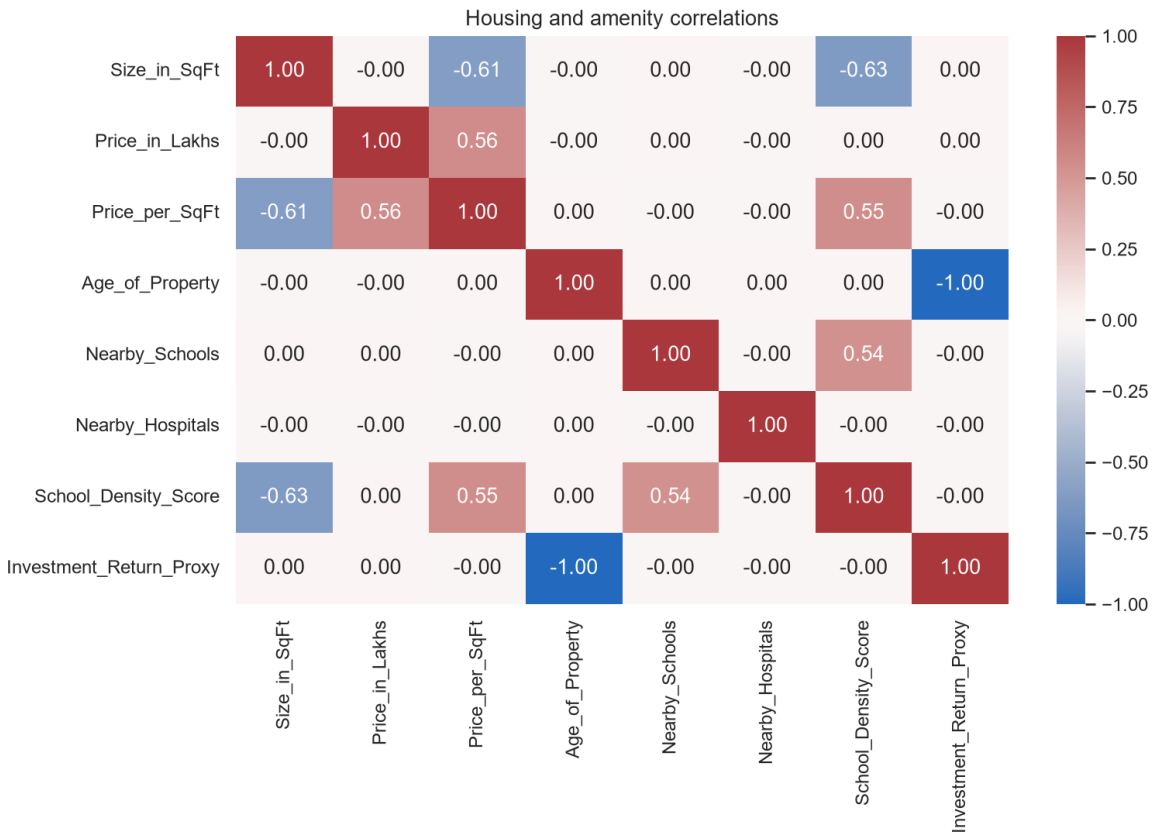


Figure 7. Correlation heatmap for selected listing fields and the investment-return proxy.

The near-perfect negative correlation between property age and the proxy is expected because age is used directly in the target formula. Other displayed associations include price with price per square foot (0.56), size with price per square foot (-0.61), and school density score with nearby schools (0.54). Correlations are descriptive and do not establish causation.

6.1. Property size and proxy-label pattern

The next view tests whether property area visibly separates the rule-generated investment-return proxy. The scatter has horizontal bands and no clear monotonic pattern across the displayed size range.



Figure 8. Property area versus investment-return proxy, coloured by the binary proxy label.

The horizontal bands are consistent with a constrained, formula-generated target. The plot should be used to inspect the proxy mechanics, not to infer that larger homes generate higher or lower realized investment returns.

INTERPRETATION Area does not show clear separation in this proxy view. Verify the target construction and replace it with observed resale and net-return outcomes before making investment claims.

7. Security indicator and proxy classifier

The security chart compares the share of listings assigned the positive proxy label between security categories. The displayed shares are nearly identical, offering little visible separation by this indicator.

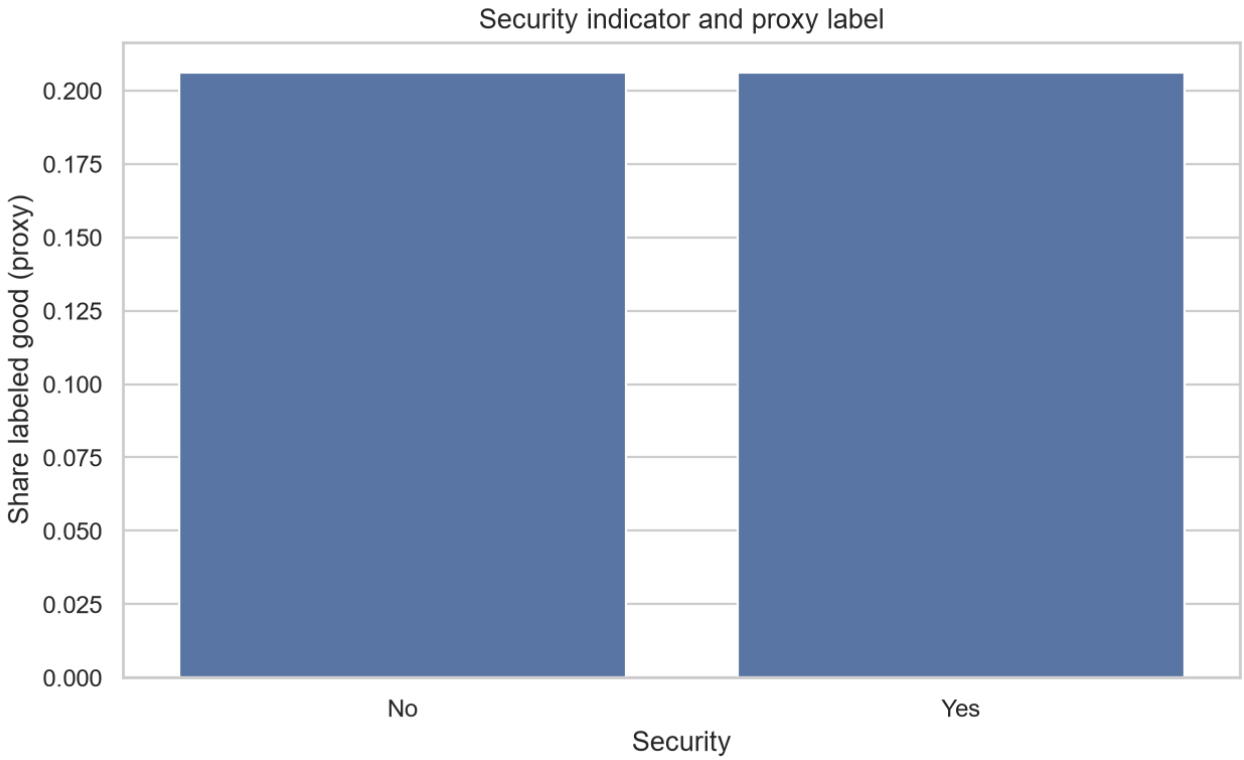


Figure 9. Security indicator and share labelled "Good Investment" by the proxy rule.

MODEL OUTPUT	HOLDOUT RESULT	INTERPRETATION
Accuracy	50.7%	Limited overall correctness for the proxy label.
Precision	20.8%	Many predicted-positive cases are not positive proxy cases.
Recall	49.4%	About half of positive proxy cases are detected.
F1 score	29.3%	Weak balance of precision and recall.
ROC AUC	0.501	Almost no ranking separation beyond chance.
Proxy price regression	MAE ₹0.00073 lakh; $R^2 \approx 1.000$	The target is formula-generated; fit does not validate resale accuracy.

MODEL DECISION Do not use the classifier to rank properties, screen actual investments or guide capital allocation. Replace the constructed target with transaction-linked resale and net-return outcomes, then retrain and validate.

8. NHB HPI forecast and model validation

The NHB RESIDEX HPI provides a separate quarterly, city-level assessment-price history. The selected Theta model forecasts 20 quarters (five years) through December 2030. This series is a city benchmark and must not be conflated with property-level asking-price data.

EVALUATION STAGE	WINDOW / SAMPLE	MAPE	PURPOSE
Rolling-origin cross-validation	12 origins · 336 city-origin forecasts	15.1%	Selected Theta among eligible candidate methods.
Temporal holdout	8 origins · 224 city-origin forecasts	16.6%	Later historical origins held out from selection; city HPI only.
Latest source index	December 2025	—	Forecast endpoint is December 2030; refresh source series before operational use.

FORECAST RESULT The selected Theta forecast improved holdout MAE by approximately 28.7% versus a last-value baseline (0.287 skill). This is historical city-index evidence; future market regimes may differ.

Forecast interpretation and constraints

- MAPE compares forecast city indices with later HPI observations; it is not error against individual property resales.
- Five-year evaluation windows overlap, so forecast errors are correlated and should not be treated as independent observations.
- The project multiplies a city forecast HPI ratio by a listing asking price. Listing capture dates and locality-level effects are unavailable, so this alignment and proportional-growth assumption is unverified.
- The empirical ratio band in project artifacts is indicative, not a calibrated prediction interval.

9. Business recommendations

The current evidence is best used to guide the next stage of analysis and data investment, not to select properties for purchase. The recommendations below translate the observed patterns and model diagnostics into practical business actions.

BUSINESS PRIORITY	RECOMMENDATION	DECISION RATIONALE
Use city analytics for discovery	Use the city-positioning chart to shortlist markets for deeper review, then compare like-for-like listings by property type, BHK, area and locality.	The chart is descriptive; it does not adjust for property mix or prove market attractiveness.
Do not operationalize the classifier	Do not use the “Good Investment” classifier to rank listings, approve purchases or allocate capital.	Holdout ROC AUC is 0.501 and F1 is 0.293 against an assumption-generated label.
Use HPI as macro context	Present the Theta forecast as a city-level scenario alongside its MAPE and forecast horizon. Keep it separate from property-specific valuation.	The HPI model forecasts a city index, not an individual asset resale price.
Shift to outcome-based evaluation	Build the next model around verified purchase and resale prices, dates, property identifiers and full ownership costs.	Observed net returns are needed to measure investment performance rather than reproduce a proxy formula.
Make assumptions visible	Show source coverage, date of last refresh, proxy assumptions, uncertainty and known gaps beside every screening output.	Transparency reduces false precision and helps decision-makers judge suitability.

Recommended decision posture

- Continue using the dashboard for descriptive market exploration and hypothesis generation.
- Require a like-for-like comparison and additional validation before presenting any city or segment as more attractive.
- Keep property-level recommendations disabled until the target is based on verified transaction-linked net returns and the model passes forward and geographic validation.

10. Limitations and decision risks

The following limitations materially constrain what can be concluded from the current analysis. They should be disclosed whenever the dashboard, forecasts or proxy scores are demonstrated or shared.

LIMITATION	WHAT IS MISSING OR UNCERTAIN	IMPACT ON DECISIONS
No realized investment outcomes	The listing file has no verified purchase prices, resale prices, transaction dates, rental income, vacancy, taxes, maintenance, financing or holding-period cash flows.	Actual ROI, net yield, capital appreciation and resale accuracy cannot be measured.
Constructed prediction target	The “Good Investment” label assumes 6% annual appreciation, subtracts 0.2 percentage points per property year and uses a 4% threshold.	Classifier performance is measured against a project rule, not observed investment success.
Weak classifier discrimination	Holdout ROC AUC is 0.501 and F1 is 0.293. The model has little ranking separation against the proxy target.	The classifier is not suitable for property ranking or investment screening.
Different market measurement bases	Listings contain asking prices; NHB RESIDEX is a city-level assessment-price index covering a separate set of cities and quarters.	These sources cannot be treated as matched property transaction histories or property-level valuations.
Limited forecast evidence	The HPI history has 51 quarters through Dec 2025; five-year evaluation windows overlap, and the empirical ratio band is not a calibrated prediction interval.	Forecast uncertainty may be understated by point estimates; results may change in new market regimes.
Representativeness and source provenance	Listing price patterns and category distributions are unusually similar in several supplied charts. Source archive version, collection process and redistribution rights need confirmation.	Market-wide generalization and public redistribution should not be assumed without further audit.
Locality factors not substantiated	Verified locality-level rent, vacancy, crime, infrastructure, accessibility and amenity quality measures are not available.	The model cannot attribute localized price or return differences to these factors.
HPI base-year maintenance	Local NHB files end in Dec 2025 and use a 2018 base year; the series must be reconciled with the newer linked base-year series before use.	Operational forecasts may be inconsistent or stale unless the index series is refreshed and reconciled.

Interpretation guardrail

Asking-price differences are descriptive, correlations do not establish causation, proxy labels are not observed outcomes, and city HPI forecasts are not individual property valuations. These outputs should not be described as financial advice or validated investment recommendations.

11. Recommended next steps

The work should proceed in the following order. The sequence prioritizes target validity first, then enrichment, rigorous validation and product controls.

PRIORITY	ACTION	WHY IT MATTERS
1 · Data	Acquire verified transaction records with property IDs, dates, locations and sale prices; document licensing.	Creates a real target for resale and return evaluation.
2 · Enrichment	Add rents, vacancy, taxes, maintenance, financing, accessibility and verified locality measures.	Enables net-return and neighborhood analysis.
3 · Validation	Use time- and geography-based holdouts; compare baselines; report uncertainty and segment errors.	Tests generalization and exposes failure areas.
4 · Operations	Refresh NHB series, log hashes/download dates, preserve raw files and monitor drift.	Maintains auditability and current inputs.
5 · Product	Label outputs as asking-price comparisons and city-index scenarios; expose assumptions.	Avoids false precision and supports informed use.

Delivery gate

Do not promote the prototype to investment-grade use until verified transaction outcomes are available, the HPI series is reconciled, and forward plus geographic holdouts demonstrate dependable performance with uncertainty reported.

12. Data, methodology and governance

SOURCE	USE IN ANALYSIS	COVERAGE / CAVEAT
India housing prices CSV (Kaggle; project attribution)	Listing attributes, asking price, location and amenities.	250,000 rows; no matched transaction dates or realized resale outcomes. Confirm archive version and redistribution rights.
NHB RESIDEX, HPI@Assessment Prices, Current-Q	City-level quarterly HPI history and forecast.	2,856 raw observations; 28 cities; 51 quarters. City benchmark, not individual property transactions.
Previous-Q NHB exports	Retained as raw provenance only.	Values differ from Current-Q for many matched city-quarter records; views are not interchangeable or averaged.

Preparation and evaluation

- Cleaning removes exact duplicates, normalizes missing tokens, coerces numeric fields, excludes nonpositive price or area, and recomputes price per square foot. The current manifest records zero exact duplicates, zero invalid-area/price removals and no missing values in modeled fields.
- Listing pipelines use imputation, numeric scaling and categorical one-hot encoding. Classification uses city-grouped cross-validation and a city holdout.
- The HPI workflow compares time-series candidates with expanding-window, 20-quarter rolling-origin forecasts; earlier origins select the method and later origins form the temporal holdout.
- NHB raw files are HTML tables saved with an .xls extension. The merged output consistently selects Current-Q. Local HPI files end in December 2025 and use a 2018 base year; reconcile with the newer linked base-year series before deployment.

Analyst conclusion

Treat this solution as an exploratory analytics prototype with useful descriptive market views, a city-index scenario and a transparent but assumption-driven screening rule. The supplied evidence does not support validated property-level investment recommendations. Decision-grade claims require transaction-linked outcomes, net-return inputs, current index reconciliation and forward validation on unseen time periods and locations.

READINESS ASSESSMENT Suitable for exploratory market analysis and hypothesis generation. Not yet suitable for investment-grade property ranking, resale valuation or financial advice.

Appendix A. Metric definitions and reproducibility

METRIC	DEFINITION
Median asking price	Median of listed property asking prices in the cleaned dataset.
Median price per sq ft	Median listing price per square foot after the project's cleaning and recomputation.
MAPE	Mean absolute percentage error for city HPI forecasts against later index observations.
ROC AUC	Area under the receiver operating characteristic curve; measures ranking separation for the proxy label.
Proxy label	Rule-generated positive/negative target based on assumed appreciation and property-age adjustment.

Reproducibility record

ITEM	RECORDED VALUE
EDA workflow	python scripts/run_eda.py
NHB source merge	python scripts/merge_nhb_hpi.py
Model training	python scripts/train.py
Forecast model artifact	artifacts/hpi_forecast_model.json
Forecast metrics artifact	reports/hpi_forecast_metrics.json
Merged NHB output SHA-256	93f72d2eb5e1554818ed091a9ece7a7feae0f9cfc820454279a8094fabe26319
Listing source SHA-256	b0032196fcf336799daf45f7fed16199ff62cf23717e6dee76911467fb1d6367

REPRODUCTION NOTE Model outputs are specific to the recorded inputs and assumptions. Regenerate reports after changing data or configuration. Source download timestamps and archive version are not embedded in the supplied listing file.

Supporting documentation: docs/methodology.md and docs/data_provenance.md. Confirm the dataset redistribution license and reconcile NHB base-year links before publishing or operational deployment.